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NBA Player Salary

An Understanding and Analysis of NBA players' Salaries



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Abstract

Player salaries in the NBA are influenced by a wide range of factors, including individual performance, experience, position, team context, and prior contract history. Understanding how these variables interact can provide insight into how teams assign market value to players and how performance translates into compensation. This has important applications in sports analytics, contract evaluation, roster construction, and salary forecasting. The goal of this project is to develop machine learning models that predict an NBA player's salary in $t+1$ using information available through season t . The $t+1$ framework was chosen because it mirrors the real decision problem faced by teams, analysts, and agents at the end of a season: using only information known through season t , estimate the player's market value for the following salary year. This setup avoids leakage, keeps the target aligned with forecasting rather than hindsight evaluation, and makes the model directly relevant to contract evaluation and roster planning.

To do this, we use a large dataset containing player statistics, advanced metrics, team-level indicators, service and experience information, and archetype-based features. We compare multiple supervised learning models including XGBoost, CatBoost, LightGBM, and blended ensemble models, across both overall and settings that are specific to a certain type of position. In addition, we look into two modeling strategies: one that includes salary-history variables for maximum predictive accuracy, and another performance-based model that excludes prior salary information in order to better isolate the relationship between on-court performance and future earnings. By evaluating these models on recent holdout seasons, this project aims to identify the strongest predictors of salary and demonstrate how machine learning can be used to estimate player value in a more systematic and interpretable way.

Data Cleaning & EDA

The dataset used for this project was assembled from multiple public sources covering the 2000–2025 NBA seasons. Source streams included Basketball-Reference (per-game, advanced, and award data), NBA.com advanced metrics, team-level outcomes, draft history, and Collective Bargaining Agreement (CBA) service buckets. These sources were merged at the player–season level to produce a single analytic file (NBADataCleanV4.csv) of roughly 10,000 rows, where each row represents one player in one season together with the target variable `next_salary` (salary in season $t+1$).

In order to clean the dataset, the first step was to reconcile player identity across eras: names spelled differently across sources, suffixes ("Jr.", "II"), and franchise relocations all required normalization before joins would behave correctly. Second, salary and contract values were standardized to nominal USD and unified across data sources so that `prev_salary`, `prev2_salary`, and `next_salary` referred to the same accounting unit. Third, advanced metrics with inconsistent historical coverage (certain hustle and tracking stats not available in earlier seasons) were either left missing for the older seasons or imputed with model-aware defaults at training time rather than at merge time.

Exploratory data analysis revealed four properties of the data that shaped downstream modeling decisions:

1. The salary distribution is heavily right-skewed. A small number of supermax and designated-veteran contracts dominate the upper tail, which means USD-MAE and USD-RMSE are both sensitive to a handful of outlier rows per season.
2. Salary is strongly path-dependent. The correlation between `prev_salary` and `next_salary` is high, which is the contract-carryover effect we set out to study and the reason the V2 modeling track was built without salary-history features.
3. Positional roles have drifted over time. Modern Point Guards produce far more points and threes than their 2000s counterparts, while Power Forwards have absorbed a stretch-shooting role. Position labels alone are not stable predictors, which motivated both position-conditioned models and archetype features.
4. Era shifts (the 2017 cap spike, the introduction of the first and second apron, repeater-tax thresholds) introduce structural breaks in the salary surface that cannot be inferred from in-sample patterns alone.

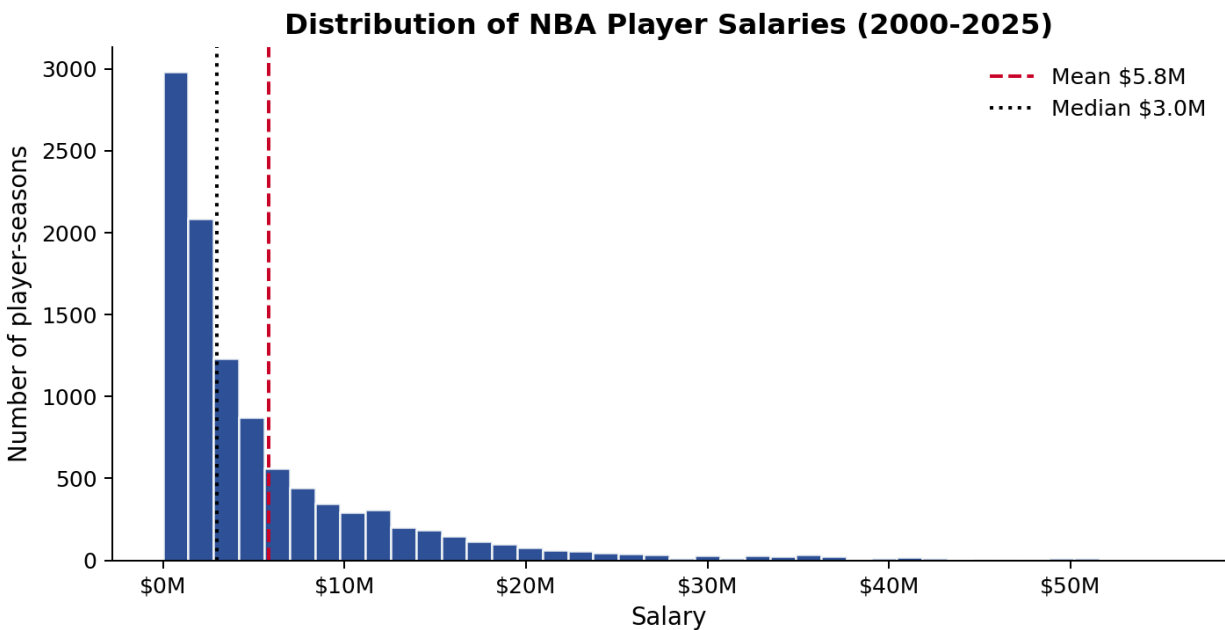


Figure 1. Distribution of NBA player-season salaries, 2000-2025 ($n = 10,469$).

As discussed above, salary is severely right-skewed. The median salary (\$2.96M) sits well below the mean (\$5.82M), and a long tail of players earn more than \$30M. This is the single most important shape in the dataset because it tells us a linear regression on raw salary will be dominated by max-contract outliers. It directly motivates two modeling choices we have already adopted: using tree-based learners (XGBoost, CatBoost, LightGBM, and a blend) that handle skew without distributional assumptions, and reporting both MAE and R-squared in dollars rather than only R-squared, since R-squared is heavily influenced by the upper tail. It also explains why naive baselines that simply carry forward last year's salary remain competitive on MAE - most of the mass sits in a narrow \$0-\$5M band where year-over-year change is small.

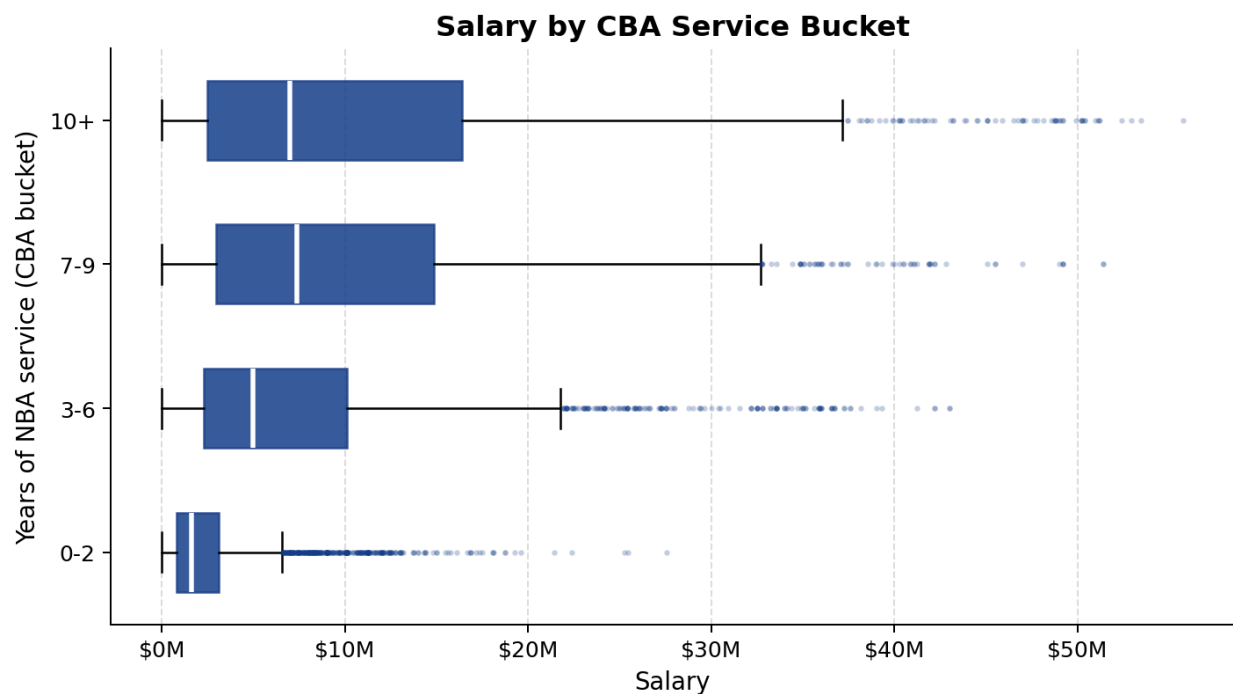


Figure 2. Salary distribution by years-of-service bucket, the structural tier that the NBA Collective Bargaining Agreement uses to set max-contract eligibility.

Additionally, as seen in Figure 2 the 0-2 year bucket is essentially flat against the rookie scale; the 3-6 bucket widens but stays compressed; the 7-9 and 10+ buckets shift sharply upward and carry almost the entire population of \$20M+ contracts. This is exactly the pattern we see in the V2 model's feature-importance output: `cba_bucket_is_0_2` and `cba_bucket_is_10_plus` consistently rank in the top five predictors for the all-players, Shooting Guard, Power Forward, and Small Forward models. The visualization confirms that years-of-service is not just a control variable but a structural cap on what a player can earn, which is why we engineered explicit CBA-bucket dummies rather than letting the models learn the rule from raw `years_of_service` alone.

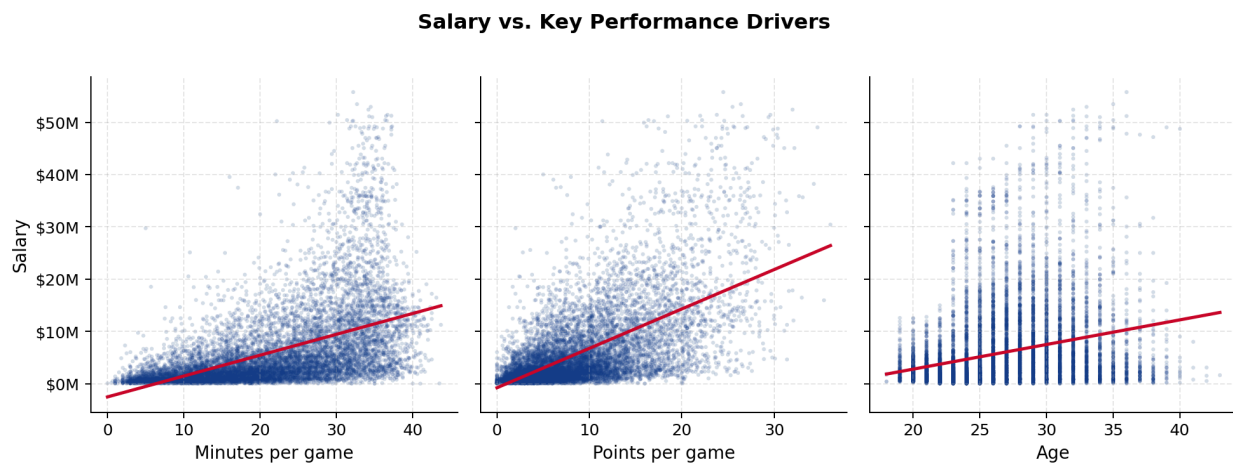


Figure 3. Salary against minutes per game, points per game, and age - the three continuous predictors that drive the V2 performance models.

When looking at salary and key performance drivers in *Figure 3*, all three panels show a clear positive trend, but with very different shapes. Minutes per game is the cleanest, near-linear signal and is the number-one feature by importance in our Shooting Guard, Power Forward, Small Forward, and Center models - role load is the closest single proxy for compensation. Points per game is positive but more dispersed at the top end, reflecting that high-volume scorers can be either max-contract stars or younger players on rookie deals. Age has a softer slope and a visible plateau in the early 30s, consistent with veterans being either max-extension holders or minimum-contract role players. Together these panels justify why a single linear model is insufficient and why tree-based models with interactions between role load, production, and career stage are the right tool for this problem.

Finally, a holdout group was set aside up front: seasons 2023–2025 were held out for final evaluation across every model variant, and no model selection or hyperparameter tuning was performed on those years.

Feature Engineering & Selection

The V2 modeling pipeline was built using two types of inputs: direct features taken from the cleaned dataset and engineered features created dynamically during training. Together, these variables were designed to capture player performance, efficiency, durability, role, and team context while avoiding direct salary leakage.

Direct Features

The primary feature set contained 65 variables used across the position-specific V2 models. These variables covered several major areas of player evaluation. First, the models included traditional box-score production, such as points, assists, rebounds, steals, blocks, turnovers, shooting attempts, shooting percentages, minutes played, and plus-minus metrics. To complement these statistics, the models also incorporated advanced metrics including usage rate, true shooting percentage, PER, BPM, VORP, Win Shares per 48 minutes, and

offensive/defensive ratings.

The feature set additionally accounted for durability and career context through variables such as games played, games missed, games started, age, height, weight, and years of NBA service. Prior recognition variables (including previous All-Star, All-NBA, and All-Defensive selections) were included to capture reputation and historical performance value.

Several team and role-based variables were also incorporated. These included team winning percentage, team net rating, conference indicators, and player archetype classifications such as lead creator, scoring wing, stretch big, and rim protector. Rolling prior-season performance measures were included as well to capture recent momentum and consistency.

Engineered Features

Beyond the direct dataset variables, the training pipeline generated 14 engineered features designed to translate basketball context into actionable signals for the model. Availability rate was calculated from games played and games missed to capture whether a player's production was dependable across the season rather than concentrated in a small sample. Starter share measured the proportion of games started and served as a cleaner role indicator than minutes alone, since two players can have similar minutes with very different responsibilities. Experience level and CBA service-bucket indicators converted raw years of service into contractual constraints, separating rookie-scale players from veterans eligible for larger extensions or maximum contracts. Draft position indicators capture pedigree and reputation effects that may influence salary even after controlling for current production. Finally, simplified positional groups and 3 year rolling trends in scoring efficiency, team success, and net rating were used to capture whether a player's role and performance were improving, declining or stable before the $t+1$ salary was observed.

Key Modeling Decisions

Two major decisions shaped the V2 framework. First, all salary-history variables were intentionally excluded. Previous salary, salary growth, and related compensation variables were removed in order to evaluate how well future salary could be predicted using basketball performance and contextual factors alone. Second, separate models were trained for each position group (PG, SG, SF, PF, and C), along with one global model across all players. This structure was designed to account for differences in positional roles and valuation trends across eras.

Feature selection combined domain expertise with empirical optimization. Variables likely to introduce leakage or redundancy were removed manually, and randomized hyperparameter search with cross-validation was used to optimize model performance using mean absolute error (MAE). The final V2 feature set remained largely consistent with V1, aside from the intentional removal of salary-history variables.

Models

XGBoost

This study utilized XGBoost to capture nonlinear feature interactions, which fits NBA salary behavior well: production, role, experience, and CBA stage interact in non-additive ways (a high-usage scorer at year 4 of service is valued very differently from the same player at year 11). It is tuned with regularization (`reg_alpha`, `reg_lambda`), `max_depth`, `learning_rate`, `n_estimators`, `subsample`, and `colsample_bytree` under a randomized search. XGBoost is also the most interpretable of the boosted models in this study: feature importance and gain attributions translate cleanly to "how much did this metric move predict salary."

Catboost

CatBoost generalizes well across grouped data (positions, archetypes) and uses ordered boosting to reduce target leakage. It was the dominant single model in V1 (selected for RefinedFeatures, Shooting Guard, and Power Forward) and won the Center model in V2. Its tolerance for default settings made it a strong baseline at the start of the project, and it remained competitive after tuning.

Blended Model (CatBoost & XGBoost)

The Blend is a convex combination of XGBoost and CatBoost predictions, with weight chosen on validation USD-MAE using a fixed grid step. It was included specifically to reduce variance on noisy, heavy-tailed targets. The Blend was selected as best for the V2 global RefinedFeatures model (XGB 0.40 / CAT 0.60) and for V2 Small Forward, and for V1 Point Guard and Small Forward, confirming that ensemble smoothing provides real gain where individual base models split residual structure.

LightGBM

LightGBM was used for its speed, allowing repeated retraining during feature engineering and randomized search without runaway runtime. Its histogram-based, leaf-wise growth captures sharp interactions that level-wise boosters miss. In V2, LightGBM was selected for Point Guard, Shooting Guard, and Power Forward, the three positions where role and minutes load matter most. Care was taken with `num_leaves` and `min_child_samples` to avoid leaf-level overfitting on the smaller position subsets.

Neural Network

A feed-forward neural network was included as a non-tree comparison point. The architecture is a small multilayer perceptron with two to three hidden layers, ReLU activations, dropout regularization, and an MAE-style loss appropriate for the heavy-tailed dollar target. Inputs are standardized; categorical archetype and CBA flags pass through as 0/1 inputs since they are one-hot encoded upstream. The neural network serves as a sanity check: on tabular

salary data with strong feature interactions and modest row counts (~10k rows total, far fewer per position), tree-based boosters consistently match or beat the MLP on holdout USD-MAE and USD-R². It is retained as a baseline rather than as a primary submission model.

For every position and for the global RefinedFeatures view, the selected model is the algorithm with the lowest USD-MAE on the 2023–2025 holdout, subject to the constraint that the cross-validation USD-MAE is not materially worse than the next-best candidate.

Results

All model results were evaluated on the 2023–2025 holdout set. Hyperparameter optimization was conducted using randomized search with six parameter combinations (`n_iter = 6`) and three-fold cross-validation (`cv = 3`). The reported metrics reflect the performance of the final selected models on previously unseen holdout data.

Position	V1 Model	V1 MAE / RMSE / R ²	V2 Model	V2 MAE / RMSE / R ²	NN MAE / RMSE / R ²
Global	CatBoost	2.80 / 4.97 / 0.822	Blend	3.55 / 5.43 / 0.816	4.11 / 6.14 / 0.765
Point Guard	Blend	4.10 / 7.33 / 0.737	Blend	4.86 / 7.55 / 0.749	6.00 / 9.01 / 0.642
Shooting Guard	CatBoost	2.73 / 4.76 / 0.760	Blend	3.36 / 5.18 / 0.791	4.86 / 7.51 / 0.560
Small Forward	Blend	2.76 / 5.10 / 0.788	LightGBM	3.31 / 4.93 / 0.810	3.98 / 6.15 / 0.705
Power Forward	CatBoost	2.96 / 4.77 / 0.853	Blend	4.75 / 7.13 / 0.728	4.63 / 7.08 / 0.731
Center	LightGBM	3.14 / 5.41 / 0.757	Blend	4.06 / 6.18 / 0.712	4.11 / 6.70 / 0.660
MAE & RMSE in \$M • holdout 2023–2025 V1 = includes salary history V2 = performance only NN = neural network baseline					

Figure 4. 2-23-2025 holdout performance for selected V1 and V2 models. MAE and RMSE are reported in millions of dollars.

Figure 4 shows that V1 produced lower MAE for every player group, confirming that salary-history variables capture meaningful contract carryover. However, V2 remained competitive, especially for guards and wings, suggesting that role, production, experience, and CBA context explain much of the salary structure even without prior salary inputs. The largest performance drop occurred for Power Forwards, where MAE increased by roughly \$1.89 million, followed by Centers, suggesting that frontcourt salaries may be more dependent on prior contract structure than guard and wing salaries.

V1: Stable Contract-Path Model (Includes Salary History)

The V1 models incorporated prior salary information, making them the strongest overall performers. The global RefinedFeaturesV1 model achieved a Mean Absolute Error (MAE) of approximately \$2.8 million and an R^2 of 0.822, indicating strong predictive power across all players.

Among the position-specific models, Power Forwards produced the strongest overall fit with an R^2 of 0.853 using CatBoost. Shooting Guards and Small Forwards also performed well, each maintaining MAE values near \$2.7 million. Point Guards proved to be the most difficult position group to model, producing higher prediction error and lower overall fit relative to the other positions. Overall, the inclusion of salary-history variables substantially improved prediction stability and accuracy across every position group.

V2: Performance-Based Model (No Salary History)

The V2 models intentionally removed all prior salary information in order to evaluate how well future salary could be predicted using only basketball-related variables. Although predictive performance declined slightly compared to V1, the models still retained strong explanatory power. The global RefinedFeatures V2 blend model achieved an MAE of approximately \$3.56 million, an RMSE of approximately \$5.50 million, and an R^2 of 0.812. These results demonstrate that future salary can still be modeled effectively using player production, role, and contextual variables alone.

Among the position-specific V2 models, Shooting Guards produced the strongest overall fit with an R^2 of 0.805, while Point Guards also maintained relatively strong predictive performance despite the removal of salary-history variables. In contrast, Power Forwards and Centers experienced the largest declines in performance, suggesting that frontcourt player valuation may depend more heavily on historical contract information.

Across both V1 and V2, LightGBM and CatBoost consistently emerged as the strongest individual model families, while blended ensemble approaches provided additional gains in several position groups. A neural network baseline achieved error levels comparable to the global LightGBM and CatBoost models; it did not outperform the tree-based approaches on any individual position group. As a result, gradient-boosted tree ensembles remained the preferred modeling framework.

Neural Network Results

The neural-network track was included as a non-tree baseline and to satisfy the project requirement for a deep-learning model. The final implementation used a Keras multilayer perceptron trained on the same performance-only feature set used by the V2 models. Compared with the initial neural-network implementation, the final class-style architecture substantially improved prediction stability through target scaling, stronger regularization, Huber loss, and prediction clipping.

As shown in *Figure 5*, the revised neural network dramatically reduced prediction error for several position groups. The largest improvement occurred for Small Forwards, where MAE decreased from approximately \$251 million to \$3.98 million. Shooting Guards also improved substantially, with MAE falling from approximately \$12.0 million to \$4.86 million. Similar gains were observed for Power Forwards, while Point Guards and Centers showed little change. Overall, the revised architecture eliminated the catastrophic outlier behavior observed in the original model and produced more consistent results across positions.

Despite these improvements, the neural network did not outperform the best gradient-boosted tree models. The final neural network achieved a global MAE of approximately \$4.11 million and an R^2 of 0.765, compared with \$3.55 million and 0.816 for the best V2 model. While competitive in some position groups (particularly Power Forwards) the neural network remained less effective than the boosted-tree approaches on this tabular salary-prediction task.

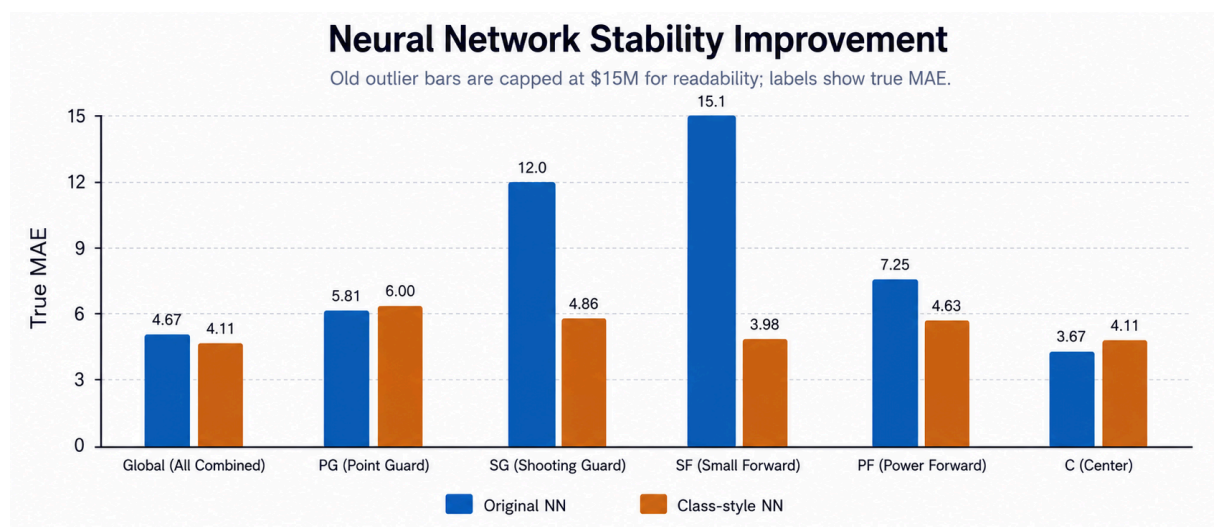


Figure 5. Neural network performance by NBA position group.

Key Drivers of Salary Prediction

Feature-importance analysis across the V2 models revealed several consistent trends. Variables tied to workload and role, particularly minutes played and previous-season minutes played, ranked among the strongest predictors in nearly every model. Career-stage and CBA-related variables also played a major role, especially indicators associated with rookie-scale contracts and veteran extensions.

Traditional production metrics remained highly influential as well. Scoring output strongly affected most offensive-oriented positions, assists were especially important for point guards, and rebounding variables were particularly influential for centers. Experience-related variables, including years of service and draft pedigree, also consistently ranked among the most predictive features.

Comparisons

Figure 6 summarizes holdout MAE across the three modeling tracks (V1, V2, and Neural Network) for each NBA position group. At the global level, the inclusion of salary-history variables in V1 improved prediction accuracy, but the improvement was smaller than expected. The RefinedFeaturesV1 model achieved an MAE of approximately \$2.80 million compared to \$3.56 million for RefinedFeaturesV2, a difference of roughly \$760,000 in average prediction error. However, the R^2 values remained very close, declining only from 0.822 in V1 to 0.812 in V2.

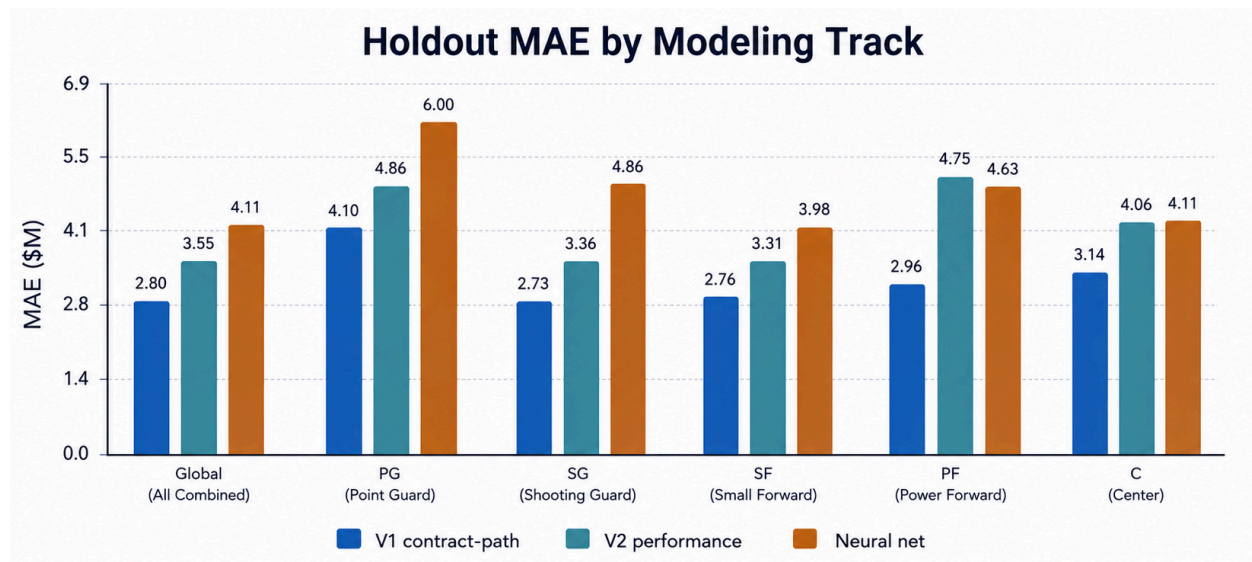


Figure 6. Modeling track performance by NBA position group.

This result suggests that salary-history variables primarily improve precision around an already strong performance-based signal rather than fundamentally changing the model's explanatory power. In other words, player production, role, and context already explain most of the underlying salary structure.

The size of the V1-to-V2 accuracy gap also varied substantially by position. Power Forwards experienced the largest decline when salary-history variables were removed, with MAE increasing by nearly \$1.9 million. Centers also showed a notable drop in performance. In contrast, Point Guards, Shooting Guards, and Small Forwards experienced smaller declines, generally between \$600,000 and \$700,000 in additional error. These patterns suggest that contracts for frontcourt players (particularly modern stretch fours and veteran bigs) are more path-dependent and tied to prior contract structure, while guard and wing salaries are more directly explained by measurable basketball production and role-based factors.

Algorithm Comparisons

The optimal model family also shifted between V1 and V2. In the V2 setting, where salary-history variables were removed, LightGBM emerged as the best-performing model for

Point Guards, Shooting Guards, and Power Forwards. This indicates that LightGBM's aggressive leaf-wise splitting strategy handled the performance-only feature space more effectively than smoother boosting approaches.

CatBoost remained especially effective for Centers, likely because of its strong handling of categorical and archetype-heavy feature interactions. Meanwhile, blended ensemble models consistently performed well at the global level and remained the strongest approach for Small Forwards across both V1 and V2, suggesting that this position benefits most from ensemble smoothing across multiple model families.

Position Difficulty

One of the most consistent findings across the experiments was the stability of position-level prediction difficulty. Shooting Guards and Small Forwards were the easiest positions to model, producing the lowest prediction error and strongest overall fit in both V1 and V2. Point Guards, Power Forwards, and Centers were consistently more difficult to predict regardless of algorithm choice. Because this ordering remained stable across model families, it likely reflects underlying characteristics of the positions themselves rather than weaknesses in any single algorithm. Differences in salary distribution, role variability, and contract structure likely contribute to this pattern.

Conclusions

This project aimed to predict NBA player salary in season $t+1$ using information available through season t under two different modeling approaches. The V1 framework included prior salary as an input, while V2 intentionally removed salary-history variables in order to measure how much future salary could be explained strictly through performance, role, durability, and contract context.

The results show that NBA salary is highly predictable even without prior salary information. The global V2 model achieved an R^2 of 0.812 and an MAE of approximately \$3.56 million on the 2023–2025 holdout set, demonstrating that a large share of salary variation can be explained through basketball-related factors alone. Across nearly every position, the strongest predictors were workload, production, career stage, and service tenure. Variables such as minutes played, points per game, assists, rebounds, and CBA service buckets consistently ranked among the most influential features.

At the same time, prior salary still contributed meaningful predictive value. Including salary-history variables in V1 reduced global MAE by roughly \$760,000 and improved accuracy across nearly every position group. This gap represents the contract-carryover effect present in the NBA salary market. Multi-year deals, veteran extensions, player options, and other contract structures are partially embedded in prior salary figures in ways that performance metrics alone cannot fully recover. As a result, V1 is the stronger framework for pure forecasting accuracy, while V2 provides a cleaner measure of whether player compensation aligns with on-court production and contextual value.

The project also showed that no single algorithm consistently outperformed the others across all positions. LightGBM emerged as the strongest model family for several V2 guard and power forward models, CatBoost performed best for centers, and blended ensembles produced the strongest global and small forward results. These findings suggest that a position-conditioned ensemble framework is more effective than applying a single model architecture uniformly across the league.

Several limitations should also be acknowledged. The NBA salary distribution is highly skewed, meaning a small number of supermax contracts disproportionately influence dollar-based error metrics. In addition, several potentially important variables (including market size, jersey sales, media exposure, and narrative value) were identified conceptually but were not incorporated into the dataset. Structural league changes, including CBA revisions and salary-cap shifts, also introduce era-specific effects that are difficult for the models to fully capture.

Overall, the findings suggest that roughly 80% of the variance in next-season NBA salary can be explained using prior-season performance, role, durability, and CBA context alone. The remaining unexplained portion reflects the path-dependent nature of NBA contracts, the exact effect that the V2 framework was designed to isolate rather than absorb through salary-history variables.